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Draft

Read Time: 13 Minutes

Title (Required)

Agentic AI: The New Digital Forensics Workhorse

Summary (Required)

Agentic AI will dramatically enhance digital forensic enabling faster, more comprehensive, deeper, and trustworthy incident response capabilities. In order to bring AI capabilities to digital forensics, product leaders must address digital chain of custody, transparency, and audit requirements.

Executive Summary (Required)

Key Findings (Required)

- Agentic AI can streamline forensic workflows without sacrificing trust. Product leaders who address “The Real” digital chain of custody requirements through systemic process and verifiable outcomes, will be in a position to develop “adoptable” industry leading Digital Forensics and Incident Response capabilities.
- Incident response providers use agents to provide detailed analysis results at machine speed. By documenting every automated step and model interaction, they ensure that the resulting evidence meets the strict Daubert and Federal Rule 901 standards for court admissibility.
- Specialized forensic analysis AI models are providing high degree of “procedural rigor.”, and 85+% accuracy rate of source attribution.

Recommendations (Required)

- Prioritize process driven Digital CoC (Chain of Custody), auditability, verification and explainability. Focus processes and tools offering “explainable” workflows and human-readable rationale through best practices and frameworks like SHAP or LIME.
- Transition from “what” happened to “how” the evidence is collected through the adoption of Digital Forensic Knowledge Graphs (DFKGs) enabling AI to visualize complex linkages and timelines.

- Overcome lack of AI trust and perceived black box process. Build trust through deployment of a transparent, explainable DFIR Tandem Model utilizing strategic interoperability between AI and Human Capital

Strategic Planning Assumption(s) (Optional)

By 2030, the traditional models of manual human dependent forensic investigation will largely be irrelevant.

Analysis (Required)

We have in recent years seen a significant increase in the use of automation and Agentic AI by threat actors, in particular Nation State affiliated Threat actor groups, to cast a significantly wider net. This has helped with the speed at which Zero Day vulnerabilities have gone from identification to being weaponized and exploited, in many cases what used to be months is now hours.

However on the defensive side the adoption of Agentic AI is lagging significantly, partially because of the regulatory and legislative requirements and partially tradition.

Background and Context (Optional)

Digital forensics and incident response (DFIR) has been somewhat lagging behind other areas of Cybersecurity services in the adoption of AI in a meaningful manner that provides real value to the end client. There is a critical path in each incident that requires speed and precision requiring a highly specialized skill set. There are many practitioners that feel the risk associated with the AI "black-box" far outweighs the benefits of AI adoption. As a result the more traditional vendors, who have been in this space for longer, and the larger service providers, have tended to focus on the use of agentic AI for the more process driven and report writing elements of their services. Although this does provide value, and does save some time it is not necessarily what should be considered within the critical path of an investigation. The primary focus should always be minimizing risk and time to recovery. Agentic AI has come a long way in a very short amount of time and will continue to do so.

For DFIR vendors to remain relevant, product leaders will need to focus the development of their capabilities on taking advantage of AI where it will be able to have the greatest impact on the investigation, forensic analysis process. Agentic AI is at the point where it can handle the far more complex tasks at scale, precision and with extreme consistency 80% to 90% of the time, that is simply not possible when done

manually.

DFIR is generally structured into 5 different areas:

- Planning and Preparation - A key component to risk limitation and mitigation is proper preparation for the inevitable incidents. The objective is to plan for who will take what action during the incident and to test this on a regular basis.
- Program and Incident Management - Focuses on the structure execution of the plan when an incident does occur in alignment with the plan. A big part of this is the effective communication at a regular cadence both internally and externally throughout the incident along with post incident activities.
- Forensic Analysis and Investigation - The largest part of the incident is always understanding the nature of the incident and how far and wide it reaches, along with the business impact. Generally this requires a very senior skill set and is one of the most time sensitive elements of an incident.
- Containment, Recovery and Remediation - In many cases, in particular with a major incident, this portion takes the most time but has a significant dependency on the effectiveness and speed at which the investigation happens.
- Legal and Compliance - Consider this the oversight over the entire incident management program, from the preparation for an incident, conduct during the incident, to the engagement with law enforcement and potential litigation implications both from a legislative oversight perspective and potential civil liability.

Each of these 5 pillars and their sub components have traditionally been very manual and labour intensive with some level of automation or use of tools being leveraged for certain aspects. Traditionally a lot of the focus for Product leaders and service providers has been a tech stack that aligns fairly closely to the data collection used by the Security Operations Center (SOC) for threat monitoring and various others to help with very specific elements of the investigation depending on its nature, with a heavy reliance on the technical expertise of individuals to be effective.

There has been some level of Agentic AI adopted to assist with more of the Softer elements of the Incident management program, such as planning and preparation and Program Management and to a lesser extent the Legal and Compliance elements. But the highly technical areas have been largely ignored or where there is some evidence of AI adoption it has once again been focused around report writing, and there has been some level of automation more aligned with the use of a Security Orchestration and Automated Response (SOAR). But this is the area where Agentic AI will have the greatest impact.

The Impact (Optional)

Agentic AI has and will continue to progress significantly in a short amount of time, but trust continues to be a hurdle for organizations to move beyond the superficial elements of the DFIR services that are helpful but do not have a meaningful impact on the client specific critical path items during the course of a security incident. The vendors that choose to embrace the adoption of Domain Specific LLM's ForensicLLM or similar

will dominate the market in the next 3 to 5 years. Client centric outcomes that focus on what is critical to the company operations such as the detection, containment, recovery, and remediation will continue to remain the core focus of most organizations and are the areas currently that are lagging behind on AI adoption. AI will not replace the Forensic investigators but rather give them scale and make these types of services more accessible to smaller organizations.

There is a likelihood you will start to see more DFIR platforms come to the forefront that will drive the process and allow smaller consultancies to compete on the larger scale, this also gives Managed Detection and Response providers the opportunity to expand their portfolio of services beyond their traditional capability without the significant cost overhead and deep bench that has traditionally been associated with DFIR retainer services.

Conclusion (Optional)

Digital Forensics and Incident Response (DFIR) is lagging behind the rest of the cybersecurity services industry in adopting Agentic AI or AI in general to enhance delivery capabilities. This is in part legacy or tradition and in part it is legislation that is lagging which is forcing the manual approach to the critical path functions within an incident. We have seen some recent movements in legislation to provide frameworks and guidelines on how evidence should be collected in order to be admissible, making it extremely difficult.

Threat actors have rapidly moved towards the adoption of large scale automation and AI for their attack approach with no regard to regulations and without any constraints which has led to a divide in both the Preemptive and response capabilities and the rapidly evolving threat actor capabilities.

The largest obstacle with the current state of agentic AI in the field is not the capabilities but rather trust, with concerns about the AI "Black Box". We are seeing startups take more risks and push the boundaries of agentic AI in forensic investigation focusing more on the Forensic Analysis & Investigation, and Response & Remediation portions and less on the process, reporting and administrative side, which is where the traditional larger vendors have ventured. This is giving startups an advantage.

How long is it going to take for product leaders to move on from this apprehension to fully embrace AI for the enhancement of all aspects of DFIR? Who is going to lead the way and dominate the market and which vendors are going to be left behind?

Much the same as what we are seeing in the software development space with vibe coding, and with the release of GPT-5.3-Codex boasting enhanced cybersecurity capabilities, there is a drive toward significantly broader adoption of AI across the cybersecurity space. Product Leaders that

are not all in on Agentic AI in the DFIR will find that their solutions will not keep up with the market demands and will become irrelevant.

Through some emerging technologies like ForensicLLM, which is still largely a research model, there is a proven approach to the use of an LLM that can achieve a very high degree of accuracy in evidentiary attribution at a scale and speed that is simply not possible by an individual or even teams. In order to overcome the trust issues, which is one of the largest barriers to the broader adoption of AI for Forensic Investigation, these models need to evolve and need to be adopted more broadly. Standards and legislative allowances for Digital CoC that align the timeline with the artifacts with the methodology used by the AI, will go a long way to bridging this divide.

It is important to note that the use of AI will not eliminate the need for the specialized investigator resources, it just shifts their focus from the large scale mundane tasks of grinding through enormous amounts of data, and detailed report writing to more of a focussed validation and understanding of the actionable evidence. Ensuring that the evidence and timelines are accurate and using the tools answer more complex questions that will assist in the training of the models long term. Forensic Investigation specialists are key to the continued enhancement and training of the models.

I would expect that there is going to be some level of government oversight and validation of platforms and tools to meet the strict standards (Daubert and Federal Rule 901 standards) required by the legal system. There is also likely to be a level of validation by cyber insurance companies as they look to mitigate their risk. We have seen for years that insurance companies have been selecting approved IR service providers that meet their standards on their ability to deliver, this will too evolve into a level of validation of AI LLM's that are being leveraged for the investigation process.

The rate at which governance and legislative standards evolve and are adopted is going to need to be rapid, it has to be. It is important that there are defined standards that are clearly adhered to, which will need constant human validation but this does not detract from the value that AI can and will provide, as the solutions and services pivot from the "what has happened" to the "how" the evidence was collected through the use of Digital Forensic Knowledge Graphs (DFKGs), giving AI the ability to visualize the timelines and linkages between artifacts and evidence that would not ordinarily and easily be evident, driving a high degree of accuracy in the evidentiary attribution.

The immediate future is going to see the need for the tandem operation where AI and human investigators work together to speed up the investigation and get to the point of containment, recovery and remediation which is the critical path for a client in the midst of a cyber incident.

Contributors (Optional)

Gartner Recommended Reading (Required)

Recommended Reading has moved out of the Google Doc canvas. Please use the Add-On or the Item Detail page to edit your Recommended Reading.

Acronym Key and Glossary Terms (Optional)

DFIR	Digital Forensics and Incident response
SHAP	Shapley Additive Explanations
LIME	Local Interpretable Model-agnostic Explanations

Evidence (Optional)

<https://dfrws.org/presentation/forensicllm-a-local-large-language-model-for-digital-forensics/>

Notes (Optional)

Concept; Internal Only — This section will not publish (Required)

Along with a Title and Summary draft, the following questions are required. Please see the offline [guidance](#) for additional information.

- 1. Client Interest:** Digital forensics and incident response is a service and capability that comes at great expense and often at the worst possible moment. This is often unplanned for and is hugely disruptive, the providers need to find an advantage that will make them significantly more efficient but still allow them to meet the chain of custody and evidentiary requirements should their be a need. The apprehension is almost always the AI blackbox that creates a blindspot for investigators when trying to determine if the evidence is accurate. This cannot be the way forward and Product leaders need to find a way of addressing this. This will result in time saved for recovery and lower costs to the end clients.

- 2. Unique:** Net New content

Content Outline; Internal Only — This section will not publish (Required)

Use this section to outline your content below. You may also link to your own outline in this section. Your outline should reflect and support the content type you plan to use and may include information beyond the minimal components noted below. We recommend that you work with your manager to determine the best structure for your outline. The minimum elements for your outline include the following:

Title: Agentic AI is the new standard for Digital Forensics

Summary:

The need for transparent, auditable models, which ensure chain of custody, and the proverbial black-box are hindering the adoption of agentic AI for digital forensics. Beyond technical capabilities creating governance and clarity in the workflow is paramount to product adoption.

Content Type: Innovation Insights

Emerging Findings:

- Agentic AI can streamline forensic workflows without sacrificing trust. Product leaders who address “The Real” digital chain of custody requirements through systemic process and

verifiable outcomes, putting product leaders in a position to develop industry leading Digital Forensics and Incident Response capabilities that operate at a significantly higher pace to keep up with the threat actors, accounting for tradition, lagging laws that are creating a greater divide between offense and defense.

- Agentic AI is evolving from a simple "detection" tool into a "methodology" engine. These systems can now explain the exact forensic recipes they follow such as memory dump triggers and artifact searches providing an auditable path from raw telemetry to final conclusion.
- Specialized models like ForensicLLM outperform general-purpose AI by maintaining a high degree of "procedural rigor." They achieve an 86.6% accuracy rate in source attribution, meaning they can point to the specific file, path, or log entry that supports every claim they make.
- The emergence of Digital Forensic Knowledge Graphs (DFKGs) allows AI to visualize complex linkages and timelines. This shifts the focus from "what" happened to "how" the evidence was found, enabling human analysts to verify the logic behind an agent's reasoning in real time.
- Top-tier incident response firms are now using these agents to provide "incident-response-level" results at machine speed. By documenting every automated step and model interaction, they ensure that the resulting evidence meets the strict Daubert and Federal Rule 901 standards for court admissibility.

Supporting Evidence:

This report is based on production metrics and technical research from the 2024–2025 cycle.

- **Performance Metrics:** Production data from AirMDR and Dropzone AI shows that agents can complete complex investigations in under 12 minutes while providing a step-by-step process report.
- **Analytic Rigor:** ForensicLLM has demonstrated a 15.79% improvement in evaluation scores (G-Eval) over base models, highlighting its superior ability to handle technical forensic terminology.
- **Knowledge Graph Utility:** Studies on DFKGs show that visualizing evidence through Knowledge Graphs significantly improves the interpretability and validity of digital testimony.
- **Legal Validation:** The proposed amendments to Federal Rule of Evidence 901 specifically address the authentication of AI-generated intelligence, making methodology-driven reporting a legal necessity.

Provisional Recommendations/Provisional Section Headers:

- a. Incident response leaders should prioritize tools that offer "explainable" workflows. Look for systems that use frameworks like SHAP or LIME to provide a human-readable rationale for every investigative decision.
 - b. Implement a "tandem operation" model where AI agents handle the massive scale of data ingestion and initial correlation, while human experts focus on validating the acquisition methodology and the final strategic interpretation.
 - c. Forensic teams must require that all AI-generated reports include a detailed evidence lineage. This should consist of deterministic unique identifiers (UIDs) and a record of the specific system locations (memory, disk, or network) where each artifact was discovered.
 - d. Ensure your AI infrastructure supports "machine-inclusive" chain of custody. Every action taken by an agent including model versions, prompts, and tool invocations must be timestamped and logged in an immutable audit trail to withstand legal scrutiny.
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- Agentic AI can streamline forensic workflows without sacrificing trust. Product leaders who address "The Real" digital chain of custody requirements through systemic process and verifiable outcomes, will put product leaders in a position to develop industry leading Digital Forensics and Incident Response capabilities that operate at a significantly higher pace to keep up with the threat actors. This approach will account for tradition and lagging laws that are creating a greater divide between offense and defense.
 - Agentic AI is evolving from a simple "detection" tool into a "methodology" engine. These systems can now explain the exact forensic recipes they follow such as memory dump triggers and artifact searches providing an auditable path from raw telemetry to final conclusion.
 - Startups that are AI first are taking greater risks, to gain an edge over their larger competitors by delving into AI driven and enhanced forensic investigation at a greater rate than the more traditional providers. Their focus has been on the critical path which really looks at the mean time to recover or restoration of service.
 - Product leaders should understand the needs and urgency of each engagement and allow for the use of Agentic AI to enable delivery teams.
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 - Implement a "tandem operation" model where AI agents handle the massive scale of data ingestion and initial correlation, while human experts focus on validating the acquisition methodology and the final strategic interpretation.
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- The legal aspect is valuable, but it is important to not that a significant amount of the time the client has no intention of going to court so evidence is irrelevant, the client is asking simple questions:
 1. What is the total impact?
 2. What was the root cause?
 3. How can we prevent this from happening again?
 4. What needs to be done to recover?
 5. How long will recovery take?
- A shift to outcomes based pricing would allow for providers to move away from a per hour pricing model, and allow for billing to be based on the outcome and phase of the project or engagement. Arguably recovery at a significantly reduced amount of time is of higher value to a client, with the containment and restoration of service at significantly reduced amount of time.
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Original SCIQA

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